

Managers Employment

INTRO:

Every MLB season some managers keep their jobs while others get fired every year and everyone assumes it's always about winning. Managers are often referred to as being on the hot seat when the season is not going well. Using data from the Sean Lahman Baseball Database we looked at three things. Doing better records actually means managers get kept. Were mid season firings mostly losing teams. And do managers who stick around longer have a better record across 2,939 manager seasons from 1901-2017. The answer to all 3 was yes. Retention goes up with a winning percentage, mid season firing where almost always on losing teams and longer tenures line up with better records.

DATA:

Our data is from the Sean Lahman Baseball database mainly the Managers, Team and People tables. The Managers.csv has one row per manager per team per season going back to 1871 with the games managed, win/lose record, and a number that tells you if a manager started the season/finished it or got swapped partway through. We merged that with the Team.csv to track franchises properly because teams move and change their names often. We only used data from 1901 onward because that's when the modern era began. Before that teams and leagues were folding left and right so managers not coming back might have just been because their team didn't exist anymore and not that they got fired. This dataset works well for what we are doing because it is a full unbroken record of the two big things we care about. win/losses and whether the same guy kept managing the same team the next year.

ANALYSIS:

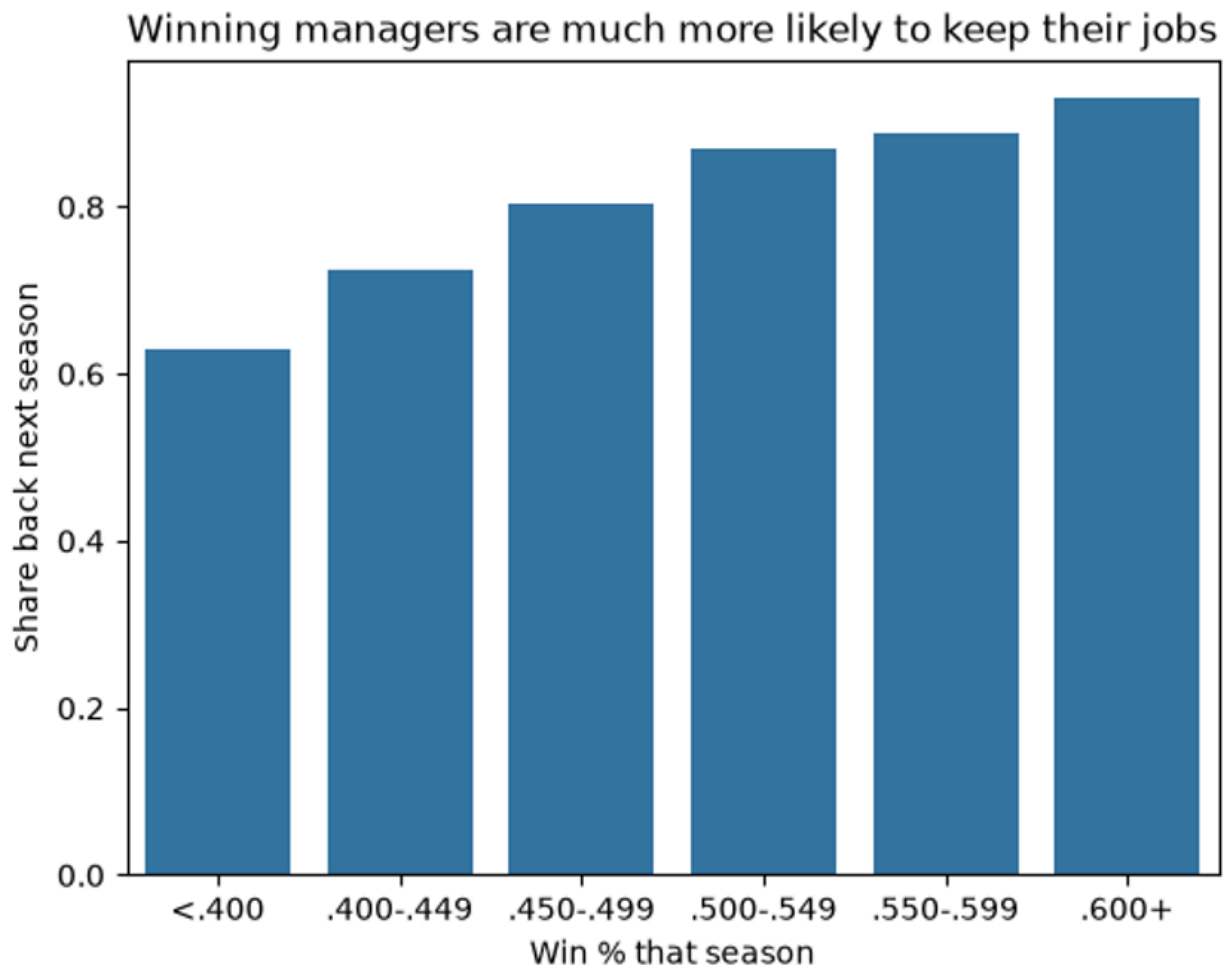
Each of our questions is basically comparing two groups. Retained vs not and replaced vs finished the season. Or looking at how two things correlate like tenure and record. We are not trying to predict anything so we went with a hypothesis test instead of building a predictive model. We used Welch's t-test to compare group means since it doesn't assume the groups are the same size or have equal variance, which matters a lot here because our groups are pretty lopsided (1959 retained managers while only 436 did not.) To answer the tenure question we used Spearman rank correlation instead of a normal linear correlation since tenure length is super skewed. Most managers only last a season or two while some stick around for decades. Basically these tests let us say whether what we found is actually bigger than what you'd expect from random chance.

RESULTS:

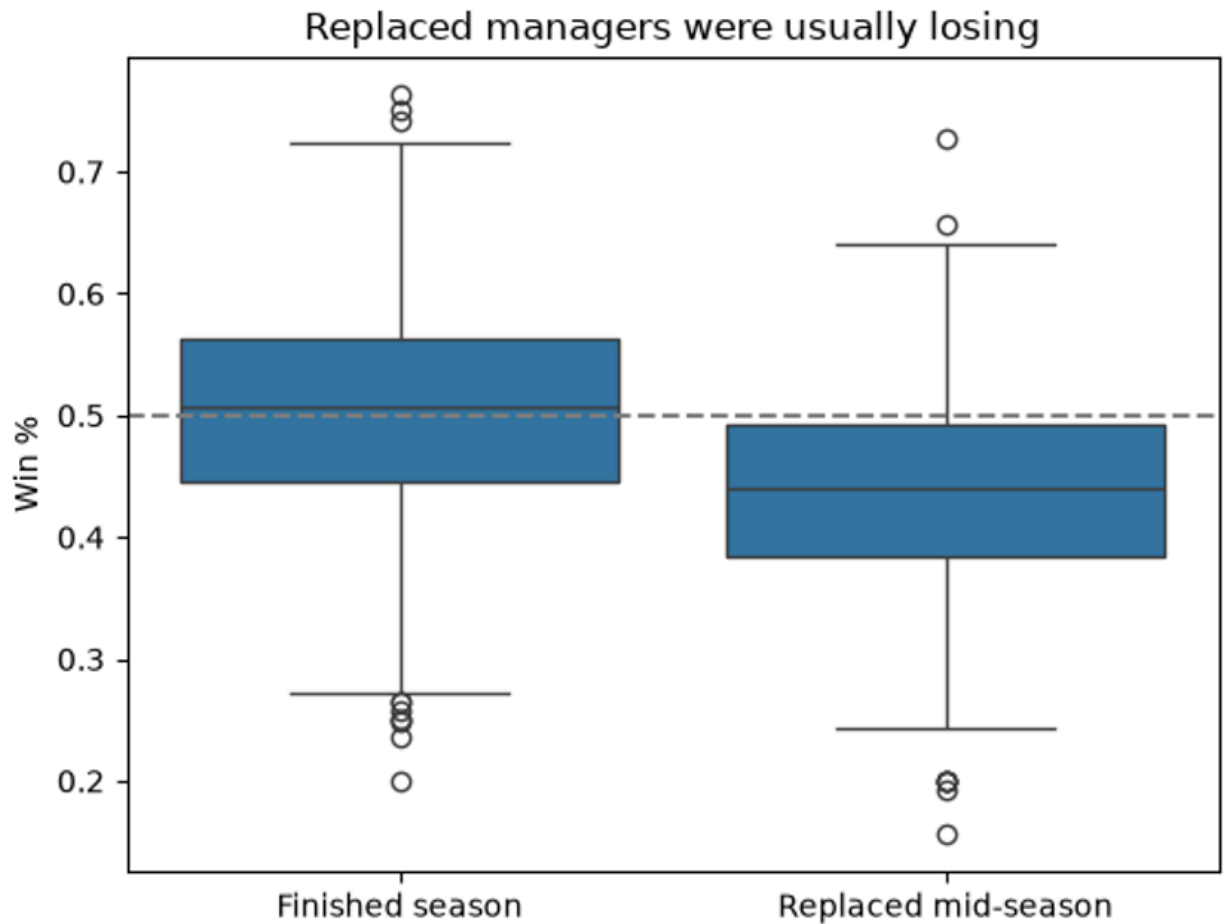
Managers who kept their jobs for the next season had won 51.2% of their games on average compared to 46.1% for the ones who got let go. This was not just some random gap between

averages. We found that retention actually climbs steadily based on your winning record. going from only 63% of sub-.400 managers coming back all the way up to 93% of managers above .600

Win % that season	Share retained
< .400	63%
.400–.449	73%
.450–.499	80%
.500–.549	87%
.550–.599	89%
.600+	93%

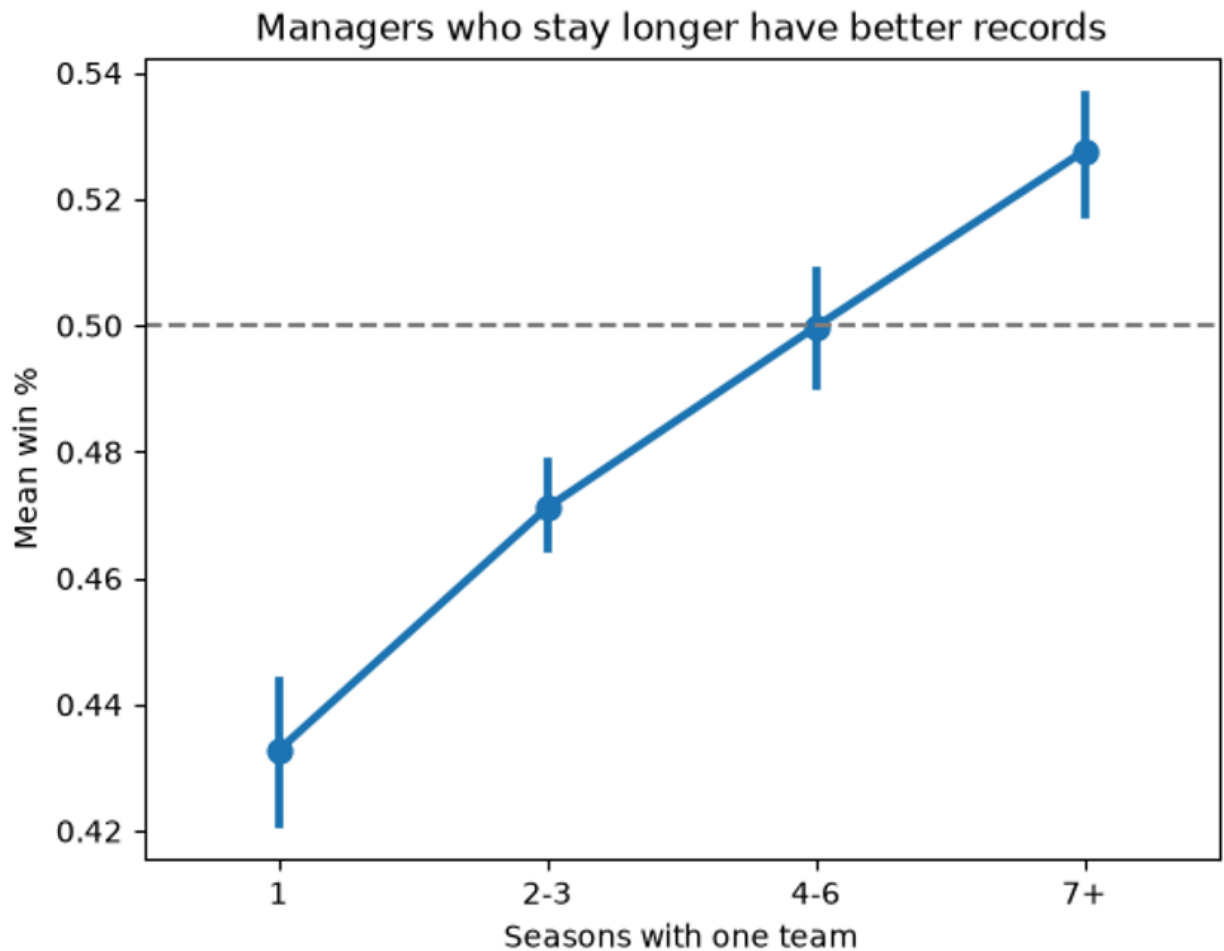


Mid-season replacements make the same point from the other direction: managers who were let go before the season ended had won just **43.8%** of their games against **50.3%** for managers who finished the year ($t = -13.3$, $p < 0.001$, $n = 357$ vs. 2,430; Figure 2). Mid-season changes were almost never a coin flip; they happened to teams that were clearly losing.



Longer tenures go with better records, too (Spearman $\rho = 0.39$, $p < 0.001$): managers in their first season with a team averaged .433, climbing to .528 for managers on a run of seven or more consecutive seasons with the same franchise (Figure 3, Table 1).

Tenure length	Mean win %	Runs (n)
1 season	.433	211
2–3 seasons	.471	341
4–6 seasons	.500	167
7+ seasons	.528	78



TECHNICAL:

Managers.csv gives one row per manager stint not per season. So a team that swapped managers mid year has multiple rows for that year. We merged in Teams to get franchID and grouped by player, year, and franchise so each manager season is one row, and we tracked the highest in season number per team year to know who actually finished the season. We cut anything before 1901 since leagues folding back then isn't the same as a manager getting fired. To check if a manager was back the next year we shifted the data forward and joined it against itself, also checking the franchise still existed so a team folding wasn't counted the same as a firing. For tenure we built runs by flagging breaks in manager, franchise, or year and dropped any run still going in 2017 since we don't know how long it would've lasted. We also dropped any season under 20 games managed since a 2 game fill in has a win percentage that's basically a coin flip. We used Welch's t-test since it doesn't assume equal variance, which matters since our groups are such different sizes. For tenure we used Spearman instead of a normal correlation since tenure length is really skewed, most managers last a season or two but a few last 20 plus years and would throw off a normal correlation. At first we joined on team ID instead of franchise ID and that was a mistake it made it look like managers weren't retained

any time their team moved or changed names so we switched to franchID. We also didn't filter by games managed at first and a couple 1 or 2 game interim managers were throwing off the averages until we added that cutoff. Something worth trying that we didn't do is a logistic regression predicting retention from win percentage plus stuff like payroll to see how much win percentage matters on its own.